

Runze Zhang

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Education

The University of Hong Kong Sept. 2026–Present

Ph.D. student, School of Computing and Data Science. Advisor: Prof. Andrew Luo.

Shenzhen Loop Area Institute, China Sept. 2026–Present

Ph.D. student, Center for Language, Intelligence and Machines.

Tongji University, China Sept. 2020–Jul. 2025

B.Eng. in Computer Science and Technology.

École Polytechnique Fédérale de Lausanne, Switzerland Aug. 2023–Feb. 2024

Semester exchange, Computer Science.

Peer-Reviewed Publications

1. **Bridging the Latency Gap with a Continuous Stream Evaluation Framework in Event-Driven Perception**

Jie Chu*, **Runze Zhang***, Chu Yang, Zongyou Yu, Zongtao Bu, Haotian Liu, Florian Röhrbein, Alois Knoll, Guang Chen, and Changjun Jiang. * *Equal contribution*.

Nature Communications, 2026. Paper · Code.

2. **Velocity Potential Field Modulation for Dense Coordination of Polytopic Swarms and Its Application to Assistive Robotic Furniture**

Lixuan Tang, David Rüegg, **Runze Zhang**, Anastasia Bolotnikova, Jan Rabaey, and Auke Ijspeert.

IEEE Robotics and Automation Letters, 2025. Paper · Code.

Manuscripts

1. **Real-Time Markerless Semantic Skeleton Tracking for Body-Language Interaction and Closed-Loop Coordination of Mobile Robotic Furniture**

Lixuan Tang, **Runze Zhang**, Minwen Mao, Paul Richard, Anastasia Bolotnikova, Emmanuel Senft, Alexandre Alahi, Jan M. Rabaey, and Auke Ijspeert.

Under review at *IEEE Transactions on Robotics*. Manuscript.

2. **Learning Task Automata for Keyframe Discovery in Robotic Manipulation**

Runze Zhang, Tongzhou Mu, Stone Tao, and Hao Su.

Preprint, February 2025. Manuscript.

Research Experience

The University of Hong Kong & Shenzhen Loop Area Institute Jun. 2026–Present

Ph.D. research · Advisor: Prof. Andrew Luo.

- Investigating how 3D world models can support perception and action planning for dexterous robotic manipulation.

Tongji University, GEAI Lab Apr. 2023–Feb. 2026

Event-driven perception · Supervisor: Prof. Guang Chen.

- Developed the STARE framework (Continuous Sampling and Latency-Aware Evaluation) and ESOT500, an event-based object-tracking dataset with 500 Hz annotations for high-dynamic motion.
- Designed Asynchronous Tracking to boost model throughput and Context-Aware Sampling to handle sparse events by adapting sampling windows and inference activation to local event density; validated latency effects in robotic table tennis.

Research Experience (continued)

UC San Diego, SU Lab

Apr. 2024–Oct. 2024

Keyframe discovery in robotic manipulation · Supervisor: Prof. Hao Su.

- Abstracted multi-stage manipulation into task automata and identified success-critical keyframes by removing or perturbing actions in expert demonstrations; tested the keyframes as supervision for manipulation policies.

EPFL, Biorobotics Lab

Oct. 2023–May 2026

3D perception and robot coordination · Supervisor: Prof. Auke Ijspeert.

- Developed a real-time, markerless 3D semantic skeleton tracking system for people and mobile furniture using multi-view RGB-D sensing, supporting body-language interaction and closed-loop furniture coordination.
- Contributed to decentralized, collision-aware coordination of dense polytopic furniture swarms, addressing deadlocks during room reconfiguration and wheelchair passage.

Teaching

Teaching Assistant, Tongji University

Sept. 2022–Jul. 2023

High-Level Language Programming I and II.

- Facilitated online Q&A sessions on C/C++ and object-oriented programming.
- Graded course projects and gave students feedback on their work.

Selected Honors

- Deep Origin Elite Scholarship, Shenzhen Loop Area Institute (RMB 25,000 monthly stipend), 2026.
- Excellent Graduate, Tongji University (top 10%), 2025.
- Excellent Student, Tongji University (top 5%), 2021, 2022, 2023.
- Excellent Undergraduate Scholarship, First Prize, Tongji University (top 5%; RMB 5,000), 2021, 2022, 2023.
- International Exchange Scholarship, First Prize, Tongji University (RMB 12,000), 2025.
- CCF Big Data & Computing Intelligence Contest, NAS Track, Second Prize (top 0.1%; RMB 10,000), 2022.

Academic Service

Volunteer, China Embodied AI Conference (CEAI), 2024 and 2025; supported media coverage and awards logistics.

Selected Technical Skills

Robots: Realman RM65-6F, OpenArm 2, OpenArm Cell, OpenArm KER, Wuji Hand 1, Inspire Hand RH5DG2.

Teleoperation and XR: Wuji Glove, PICO 4 Ultra.

Cameras: DAVIS346, DVSync, MS Azure Kinect, Vicon MoCap, ZED Mini.

Programming and ML: C/C++, Python, MATLAB, Assembly language, Verilog HDL, PyTorch.

Systems and frameworks: ROS 2, Docker, Kubernetes, Git, Vue, Spring Boot, Electron.

Tools: MS Office Suite (Word, Excel, PowerPoint), LaTeX, Markdown, Adobe Photoshop, Adobe Illustrator.

Languages: Chinese (native), English (IELTS band 7), French (A1).